软件分析

南京大学
计算机科学与技术系
程序设计语言与静态分析研究组
李樾 谭添

Static Program Analysis

Pointer Analysis Foundations (I)

Nanjing University

Tian Tan

2021

Contents

1. Pointer Analysis: Rules
2. How to Implement Pointer Analysis
3. Pointer Analysis: Algorithms
4. Pointer Analysis with Method Calls

Contents

1. **Pointer Analysis: Rules**
2. How to Implement Pointer Analysis
3. Pointer Analysis: Algorithms
4. Pointer Analysis with Method Calls

Pointer-Affecting Statements

New	`x = new T()`
Assign	`x = y`
Store	`x.f = y`
Load	`y = x.f`
Call	`r = x.k(a, …)`

First focus on these statements
(suppose the program has just one method)

Will come back to this in
pointer analysis with method calls

Domain and Notations

Variables:	$x, y \in \mathrm{V}$
Fields:	$f, g \in \mathrm{F}$
Objects:	$o_i, o_j \in \mathrm{O}$
Instance fields:	$o_i.f, o_j.g \in \mathrm{O} \times \mathrm{F}$
Pointers:	$\mathrm{Pointer} = \mathrm{V} \cup (\mathrm{O} \times \mathrm{F})$
Points-to relations:	$pt : \mathrm{Pointer} \rightarrow \mathcal{P}(\mathrm{O})$

- $\mathcal{P}(\mathrm{O})$ denotes the powerset of O
- $pt(p)$ denotes the points-to set of p

Rules

Kind	Statement	Rule
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$
Assign	`x = y`	$\dfrac{o_i \in pt(y)}{o_i \in pt(x)}$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(y)}{o_j \in pt(o_i.f)}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(o_i.f)}{o_j \in pt(y)}$

Rules

Kind	Statement	Rule
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$ ← unconditional
Assign	`x = y`	$\dfrac{o_i \in pt(y)}{o_i \in pt(x)}$ ← premises ← conclusion
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(y)}{o_j \in pt(o_i.f)}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(o_i.f)}{o_j \in pt(y)}$

Rule: New

$$\frac{}{o_i \in pt(x)}$$

Conclusion

o_i

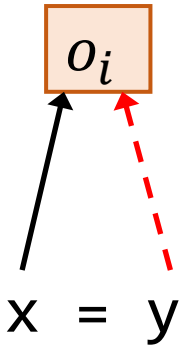
```
i: x = new T()
```

Rule: Assign

$$\frac{o_i \in pt(y)}{o_i \in pt(x)}$$

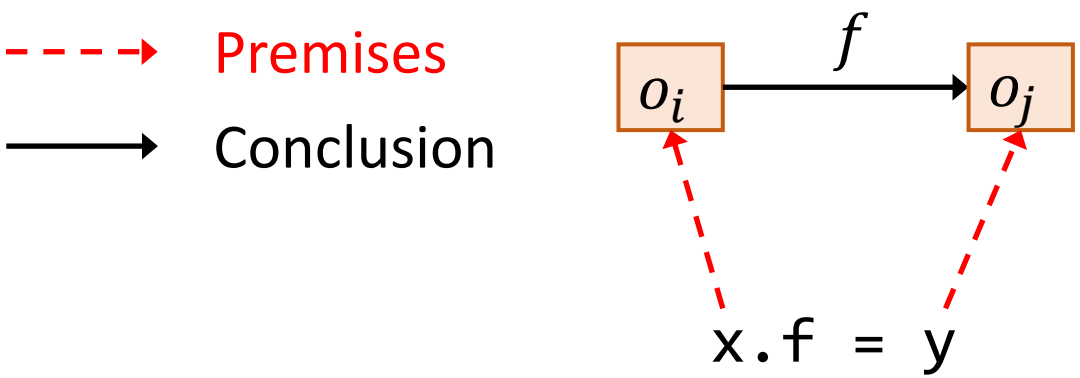
Premises

Conclusion



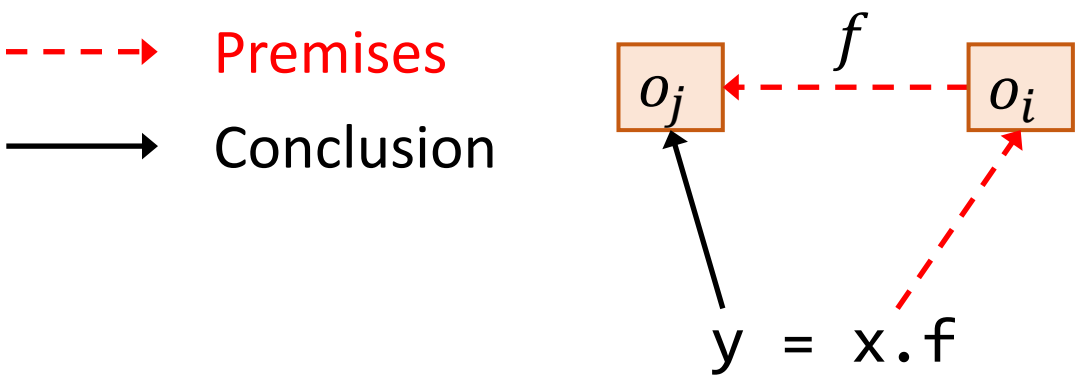
Rule: Store

$$\frac{o_i \in pt(x),\ o_j \in pt(y)}{o_j \in pt(o_i.f)}$$



Rule: Load

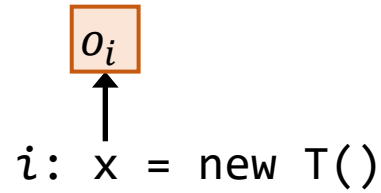
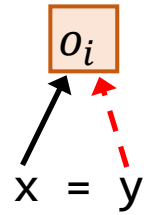
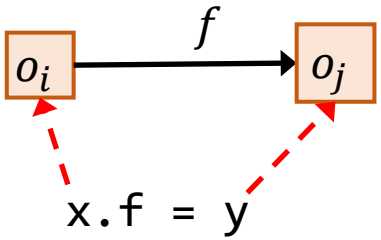
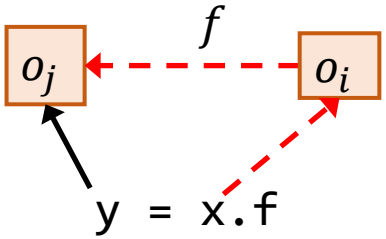
$$\frac{o_i \in pt(x),\ o_j \in pt(o_i.f)}{o_j \in pt(y)}$$



Rules

Premises

Conclusion

Kind	Rule	Illustration
New	$\dfrac{}{o_i \in pt(x)}$	
Assign	$\dfrac{o_i \in pt(y)}{o_i \in pt(x)}$	
Store	$\dfrac{o_i \in pt(x),\ o_j \in pt(y)}{o_j \in pt(o_i.f)}$	
Load	$\dfrac{o_i \in pt(x),\ o_j \in pt(o_i.f)}{o_j \in pt(y)}$	

Contents

1. Pointer Analysis: Rules
2. **How to Implement Pointer Analysis**
3. Pointer Analysis: Algorithms
4. Pointer Analysis with Method Calls

Our Pointer Analysis Algorithms

- A complete whole-program pointer analysis
- Carefully designed for understandability
- Easy to follow and implement

How to Implement Pointer Analysis?

- Essentially, pointer analysis is to **propagate** points-to information among pointers (variables & fields)

Kind	Statement	Rule
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$

How to Implement Pointer Analysis?

- Essentially, pointer analysis is to **propagate** points-to information among pointers (variables & fields)

Kind	Statement	Rule
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$
Assign	`x = y`	$\frac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$
Store	`x.f = y`	$\frac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$
Load	`y = x.f`	$\frac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$

Pointer analysis as solving a system of inclusion constraints for pointers

Referred as *Andersen-style analysis**

∗ Lars Ole Andersen, 1994. "*Program Analysis and Specialization for the C Programming Language*". Ph.D. Thesis. University of Copenhagen.

How to Implement Pointer Analysis?

- Essentially, pointer analysis is to **propagate** points-to information among pointers (variables & fields)

Kind	Statement	Rule
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$

难点在于

Key to implementation: when $pt(x)$ is **changed**, **propagate** the **changed part** to the **related pointers** of x

How to Implement Pointer Analysis?

- Essentially, pointer analysis is to **propagate** points-to information among pointers (variables & fields)

Kind	Statement	Rule
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$

Solution

- We use a **graph** to connect related pointers
- When $pt(x)$ changes, propagate the changed part to x's **successors**

Key to implementation: when $pt(x)$ is **changed**, **propagate** the **changed part** to the **related pointers** of x

Pointer Flow Graph (PFG)

Pointer flow graph of a program is a *directed graph* that expresses how objects flow among the pointers in the program.

Pointer Flow Graph (PFG)

Pointer flow graph of a program is a *directed graph* that expresses how objects flow among the pointers in the program.

- Nodes: $\text{Pointer} = \text{V} \cup (\text{O} \times \text{F})$

  A node n represents *a variable* or *a field of an abstract object*

- Edges: $\text{Pointer} \times \text{Pointer}$

  An edge $x \rightarrow y$ means that the objects pointed by pointer x *may flow to* (and also be pointed to by) pointer y

Pointer Flow Graph: Edges

- PFG edges are added according to the statements of the program and the corresponding rules

Kind	Statement	Rule
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$

Pointer Flow Graph: Edges

- PFG edges are added according to the statements of the program and the corresponding rules

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$
Store	`x.f = y`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$	$o_i.f \leftarrow y$
Load	`y = x.f`	$\dfrac{o_i \in pt(x),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$	$y \leftarrow o_i.f$

Pointer Flow Graph: An Example

Program

初始化集合

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;     ①
c.f = a;   ②
d = c;     ③
c.f = d;   ④
e = d.f;   ⑤
```

Pointer flow graph

- Variable node **v**
- Instance field node o_i**.f**

Pointer Flow Graph: An Example

Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node (v)
- Instance field node (o_i.f)

Pointer Flow Graph: An Example

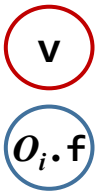
Program

$(o_i \in pt(c), o_i \in pt(d))$

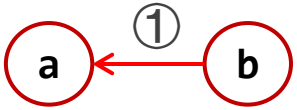
```
a = b;     ①
c.f = a;   ②
d = c;     ③
c.f = d;   ④
e = d.f;   ⑤
```

Pointer flow graph

- Variable node
- Instance field node



初始化图



Pointer Flow Graph: An Example

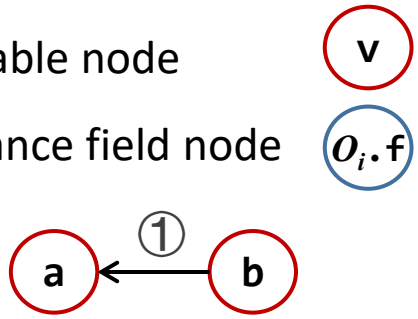
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

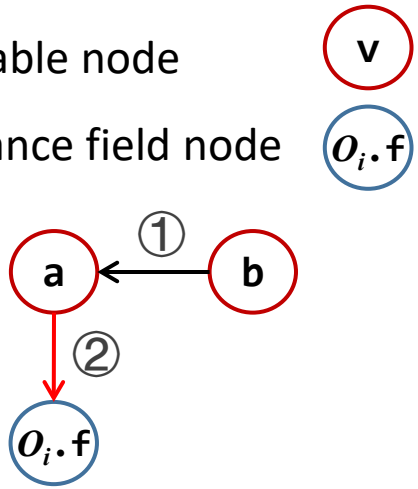
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

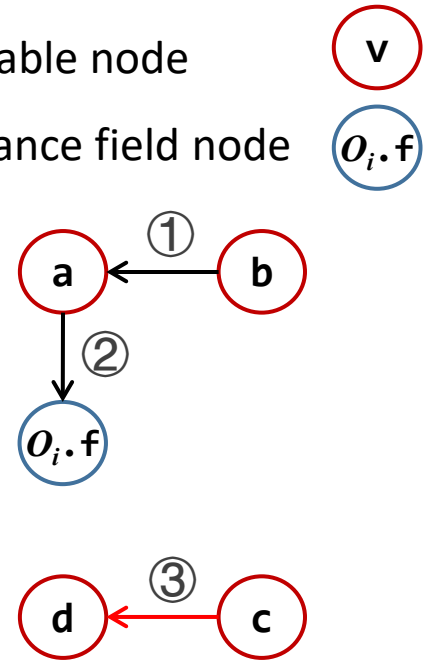
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

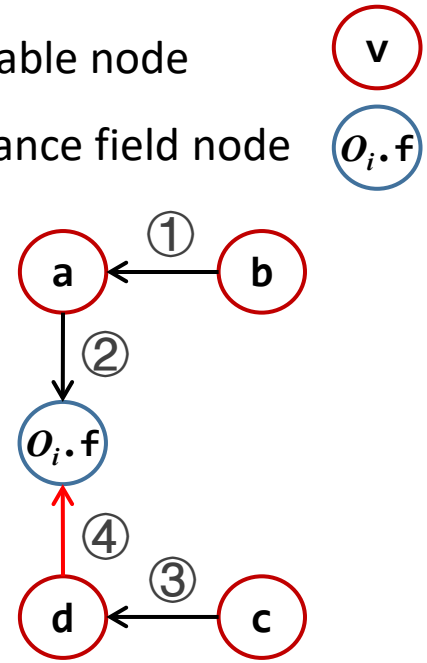
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

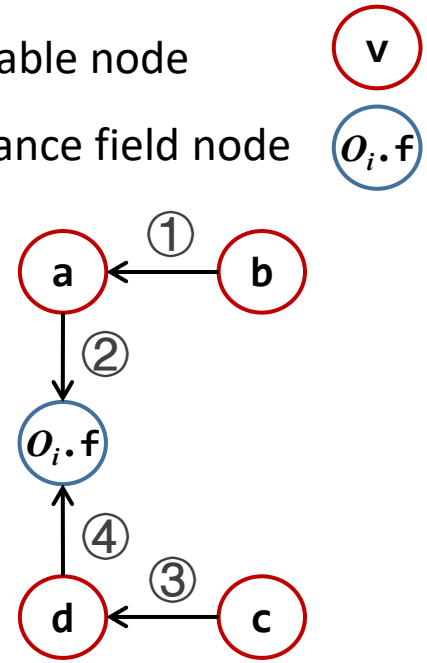
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤ ?
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

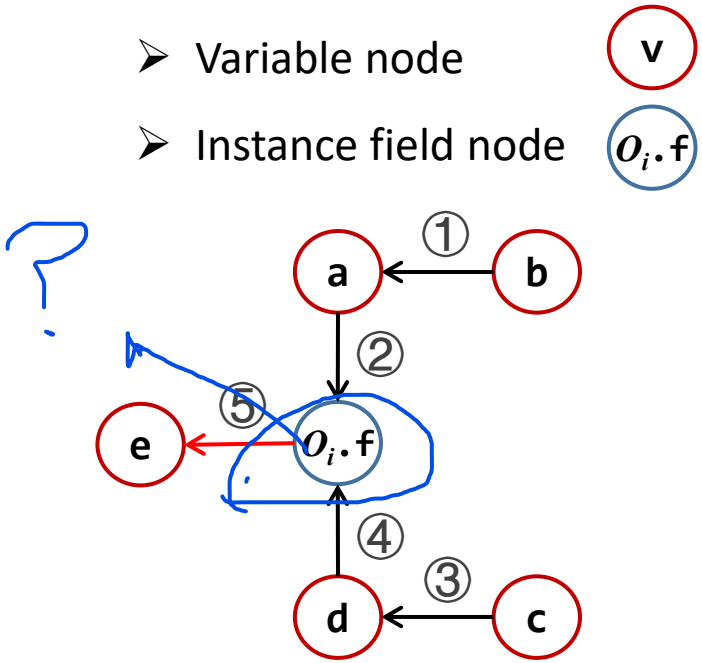
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



c.f或者d.f是不对的
c.f不表示指向一个对象

在Pointer flow graph中，图中节点只能是
1）变量
2）instance field

Pointer Flow Graph: An Example

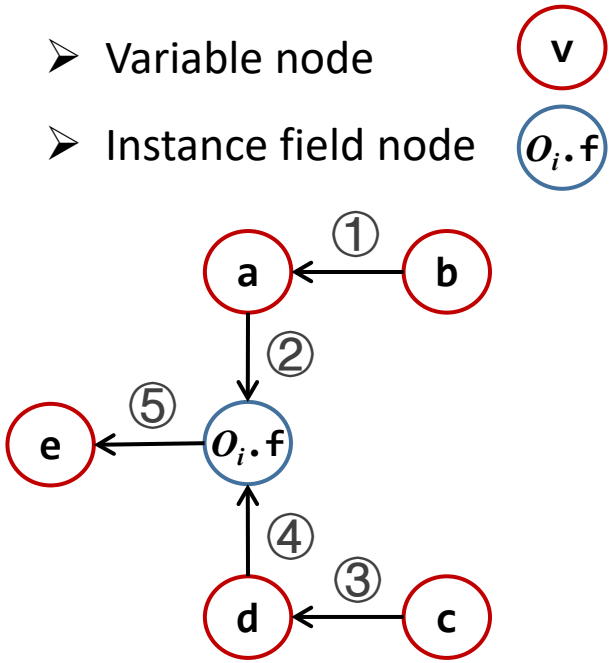
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



Pointer Flow Graph: An Example

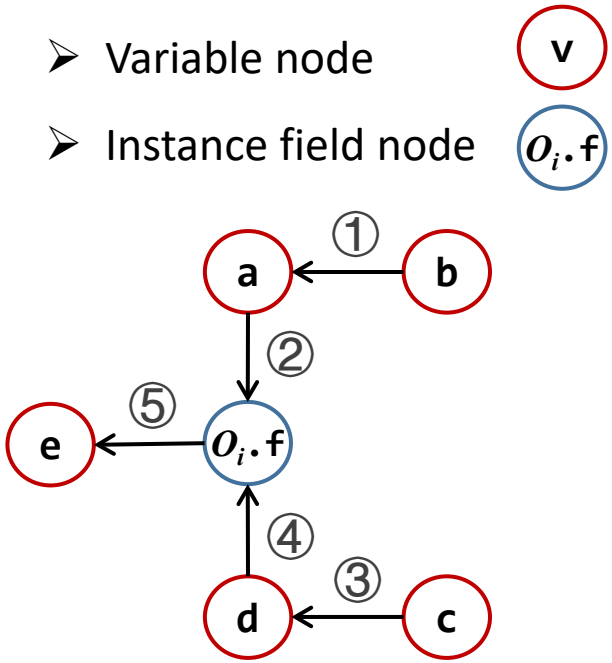
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;     ①
c.f = a;   ②
d = c;     ③
c.f = d;   ④
e = d.f;   ⑤
```

Pointer flow graph

- Variable node
- Instance field node



With PFG, pointer analysis can be solved by computing ***transitive closure*** of the PFG

传递闭包

Pointer Flow Graph: An Example

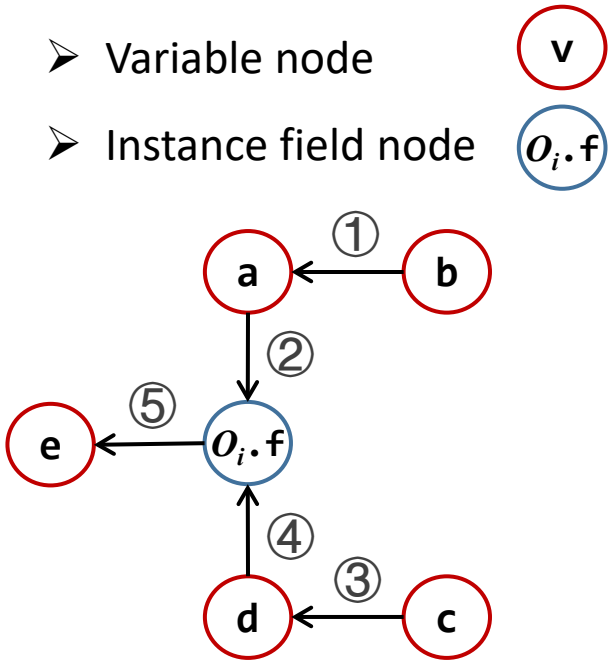
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



With PFG, pointer analysis can be solved by computing ***transitive closure*** of the PFG

E.g, **e** is reachable from **b** on the PFG, which means that the objects pointed by **b** may flow to and also be pointed by **e**

Pointer Flow Graph: An Example

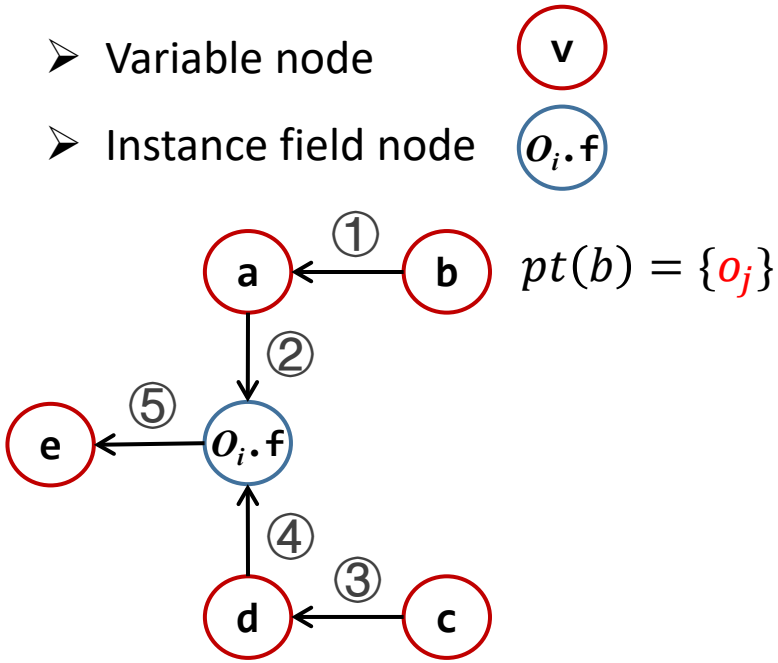
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
j: b = new T();
   a = b;      ①
   c.f = a;    ②
   d = c;      ③
   c.f = d;    ④
   e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



With PFG, pointer analysis can be solved by computing ***transitive closure*** of the PFG

E.g, **e** is reachable from **b** on the PFG, which means that the objects pointed by **b** may flow to and also be pointed by **e**

Pointer Flow Graph: An Example

Program

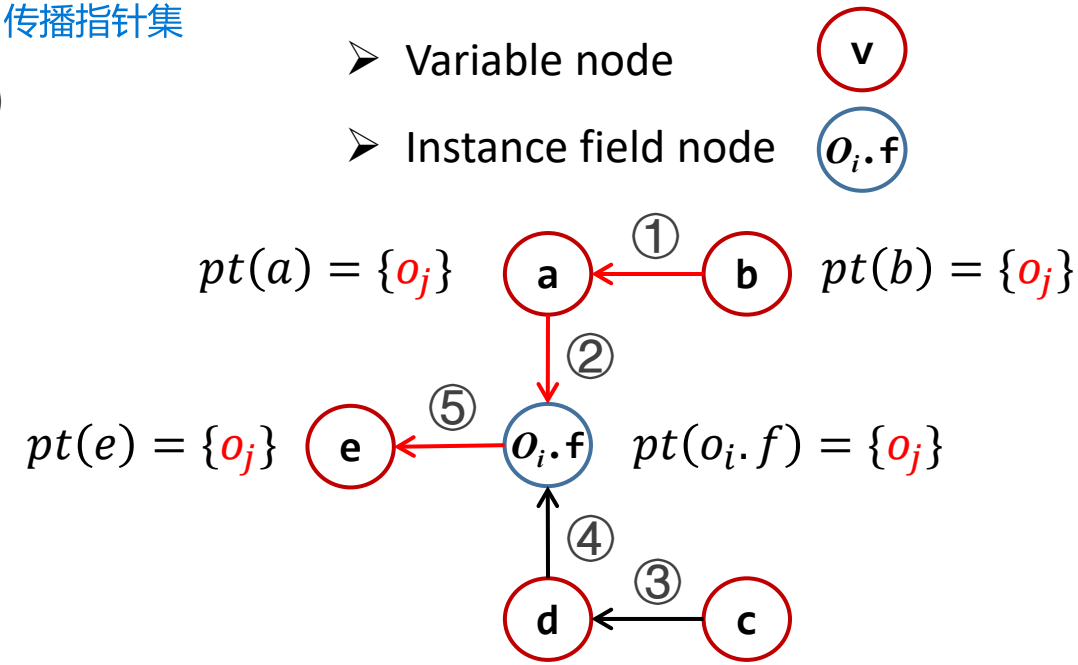
传播指针集

$(o_i \in pt(c), o_i \in pt(d))$

```
j: b = new T();
   a = b;      ①
   c.f = a;    ②
   d = c;      ③
   c.f = d;    ④
   e = d.f;    ⑤
```

Pointer flow graph

- Variable node
- Instance field node



With PFG, pointer analysis can be solved by computing ***transitive closure*** of the PFG

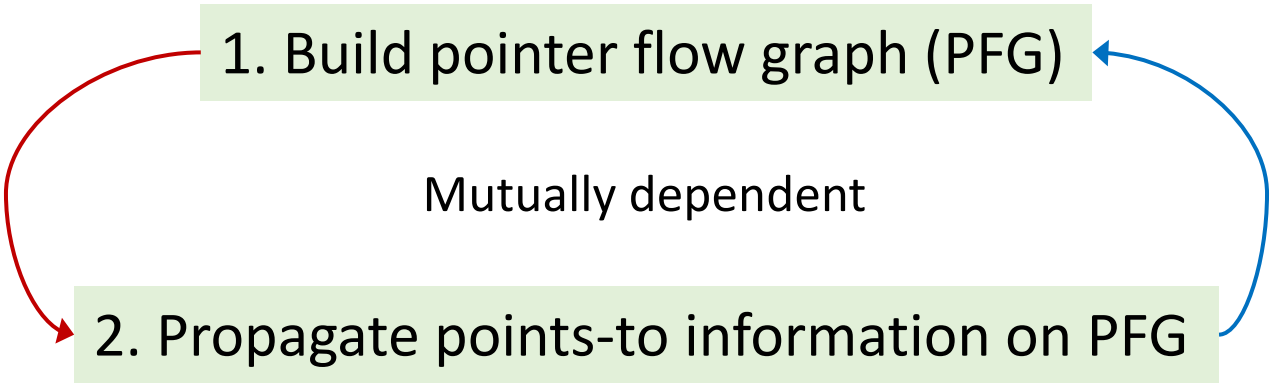
E.g, **e** is reachable from **b** on the PFG, which means that the objects pointed by **b** may flow to and also be pointed by **e**

Implementing Pointer Analysis

1. Build pointer flow graph (PFG)

2. Propagate points-to information on PFG

Implementing Pointer Analysis



Implementing Pointer Analysis

1. Build pointer flow graph (PFG)

Mutually dependent

2. Propagate points-to information on PFG

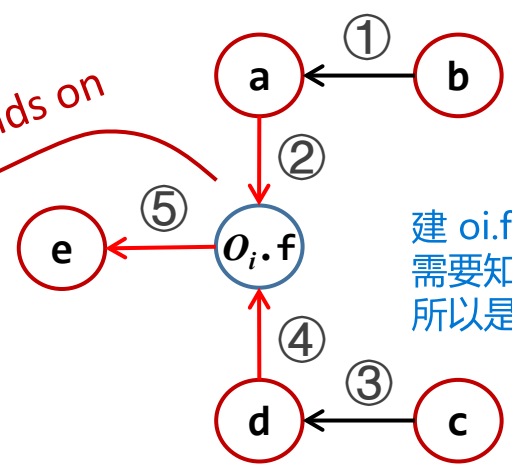
Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

depends on

Pointer flow graph



建 oi.f时 就是分析 c.f时
需要知道 oi属于c才能建立图，
所以是不断更新的

Implementing Pointer Analysis

1. Build pointer flow graph (PFG)

Mutually dependent

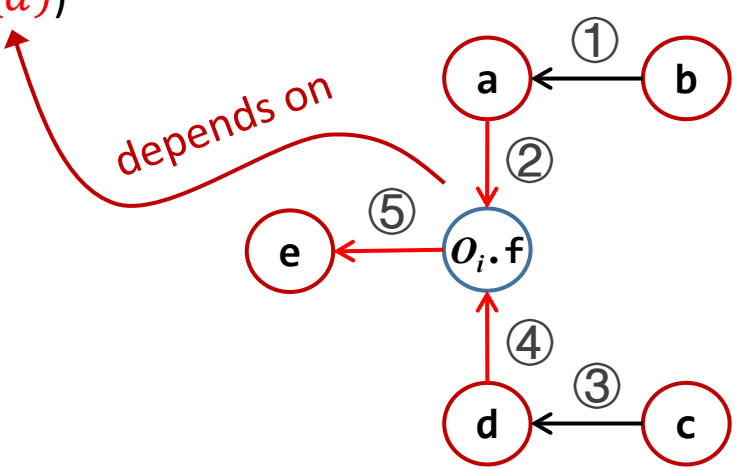
2. Propagate points-to information on PFG

Program

$(o_i \in pt(c), o_i \in pt(d))$

```
a = b;      ①
c.f = a;    ②
d = c;      ③
c.f = d;    ④
e = d.f;    ⑤
```

Pointer flow graph



PFG is dynamically **updated** during pointer analysis

Contents

1. Pointer Analysis: Rules
2. How to Implement Pointer Analysis
3. **Pointer Analysis: Algorithms**
4. Pointer Analysis with Method Calls

Pointer Analysis: Algorithms

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable `x` **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S	Set of statements of the input program
WL	Work list
PFG	Pointer flow graph

Pointer Analysis: Algorithms

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

main algorithm

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S	Set of statements of the input program
WL	Work list
PFG	Pointer flow graph

Pointer Analysis: Algorithms

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable `x` **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

main algorithm

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S	Set of statements of the input program
WL	Work list
PFG	Pointer flow graph

Worklist (WL)

- Worklist contains the points-to information to be processed
  - $WL \subseteq \langle \text{Pointer}, \mathcal{P}(\text{O}) \rangle^*$

- Each worklist entry $\langle n, pts \rangle$ is a pair of pointer n and points-to set pts, which means that pts should be propagated to $pt(n)$
  - E.g., $\left[\langle x, \{o_i\} \rangle, \langle y, \{o_j, o_k\} \rangle, \langle o_j.f, \{o_l\} \rangle \ldots\right]$

Main Algorithm

```
Solve(S)
  WL = [ ], PFG = {}
  foreach i: x = new T() ∈ S do
    add ⟨x, {o_i}⟩ to WL

  foreach x = y ∈ S do
    AddEdge(y, x)

  while WL is not empty do
    remove ⟨n, pts⟩ from WL
    Δ = pts – pt(n)
    Propagate(n, Δ)
    if n represents a variable x then
      foreach o_i ∈ Δ do
        foreach x.f = y ∈ S do
          AddEdge(y, o_i.f)
        foreach y = x.f ∈ S do
          AddEdge(o_i.f, y)
```

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Initialize the analysis

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Initialize the analysis

Add assign edges to PFG

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then** ①
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

① Do nothing if $s \rightarrow t$ is already in PFG

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then** ①
    add $s \rightarrow t$ to PFG ②
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

① Do nothing if $s \rightarrow t$ is already in PFG

② **Add PFG edge**

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then** ①
    add $s \rightarrow t$ to PFG ②
    if $pt(s)$ is not empty **then** ③
      add $\langle t, pt(s)\rangle$ to WL

① Do nothing if $s \rightarrow t$ is already in PFG

② **Add PFG edge**

③ **Ensure** every object pointed by s is also pointed by t

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Process the entries in WL

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

remove $\langle x, \{o_1, o_3\}\rangle$ from WL

$pt(x) = \{o_1, o_2\}$

$\Delta = pts - pt(x)$
$= \{o_1, o_3\} - \{o_1, o_2\}$
$= \{o_3\}$

Propagate($x, \{o_3\}$)

Process the entries in WL

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Propagate(n, pts)
  if pts is not empty **then** ①
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

① Do nothing if pts is empty

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Propagate(n, pts)
  if pts is not empty **then** ①
    $pt(n) \cup= pts$ ②
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

① Do nothing if pts is empty

② **Propagate pts to points-to set of n**

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Propagate(n, pts)
  if pts is not empty **then** ①
    $pt(n) \cup= pts$ ②
    foreach $n \rightarrow s \in PFG$ **do** ③
      add $\langle s, pts\rangle$ to WL

① Do nothing if pts is empty

② **Propagate pts to points-to set of n**

③ Propagate pts (**the changed part**) to n's successors on PFG

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\overline{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Differential Propagation

Solve(S)

…

while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…

Why?

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

```
Solve(S)
    …
    while WL is not empty do
        remove ⟨n, pts⟩ from WL
        Δ = pts − pt(n)
        Propagate(n, Δ)
        …
```

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

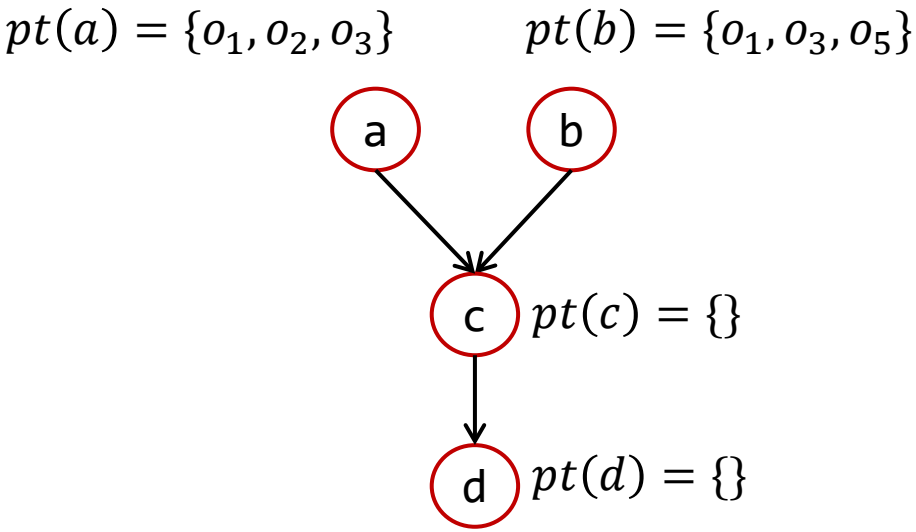
while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…



PFG

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

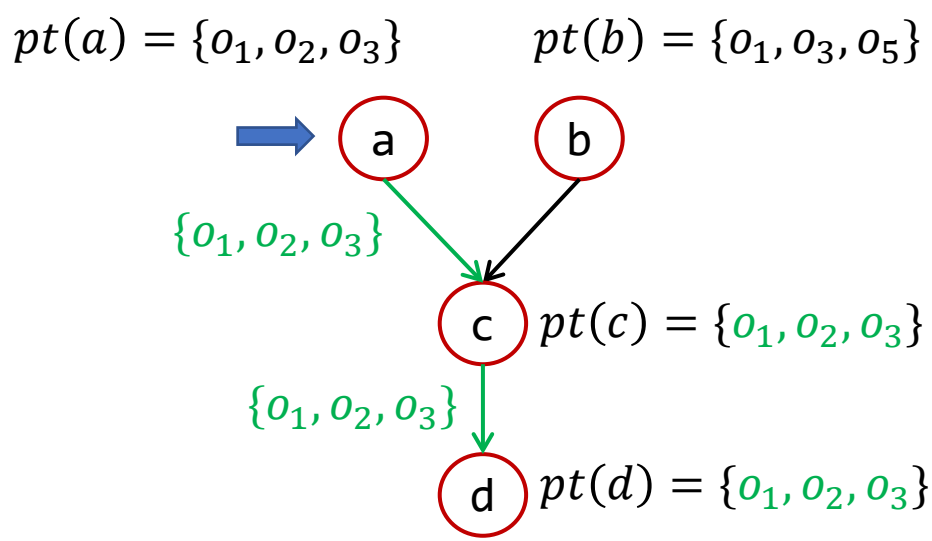
while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…



PFG

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

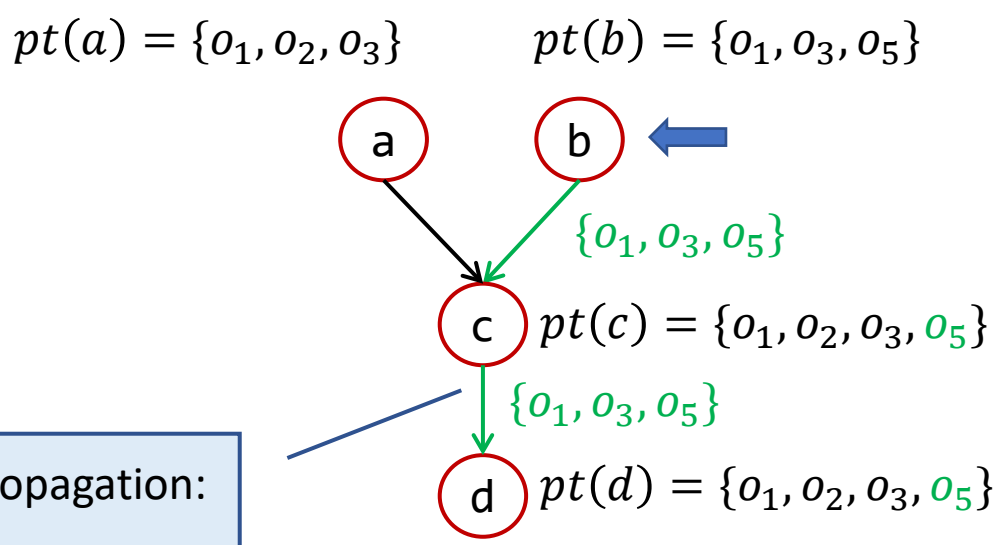
while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…



Direct propagation:

PFG

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

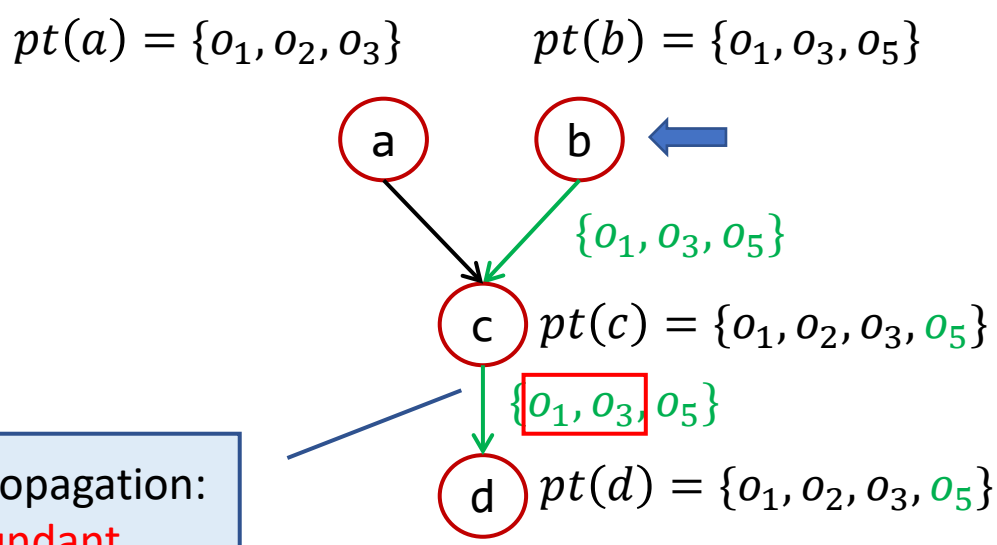
while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…



Direct propagation: redundant

PFG

Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

while WL is not empty **do**

remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…

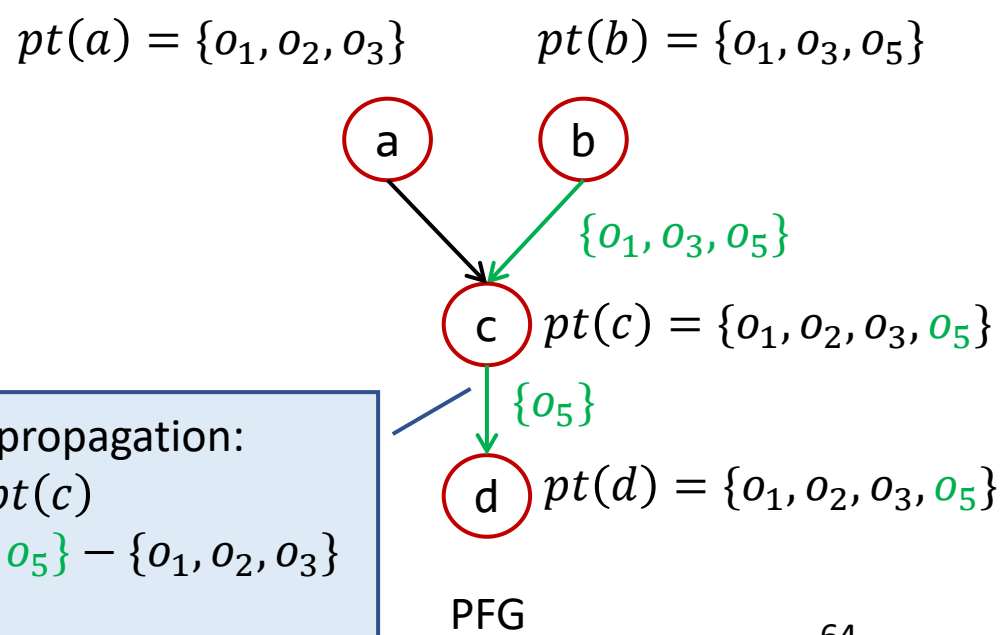
In practice, Δ is usually small compared with the original set, so propagating only the new points-to information (Δ) improves efficiency

Differential propagation:

$\Delta = pts - pt(c)$

$= \{o_1, o_3, o_5\} - \{o_1, o_2, o_3\}$

$= \{o_5\}$



Differential Propagation

- ***Differential propagation*** is employed to avoid propagation and processing of redundant points-to information
- **Insight**: existing points-to information in $pt(n)$ have already been propagated to n's successors, and ***no need*** to be propagated again

Solve(S)

…

while WL is not empty **do**

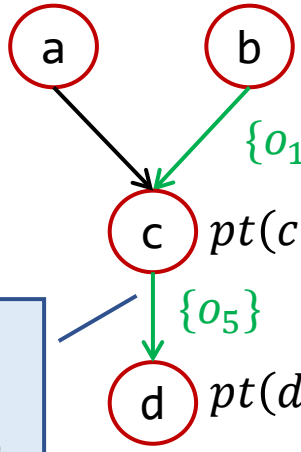
remove $\langle n, pts \rangle$ from WL

$\Delta = pts - pt(n)$

Propagate(n, Δ)

…

$pt(a) = \{o_1, o_2, o_3\}$ $pt(b) = \{o_1, o_3, o_5\}$



$pt(c) = \{o_1, o_2, o_3, o_5\}$

$pt(d) = \{o_1, o_2, o_3, o_5\}$

PFG

Besides, Δ is also important for efficiency when handling stores, loads, and method calls, as explained later

Differential propagation:

$\Delta = pts - pt(c)$

$= \{o_1, o_3, o_5\} - \{o_1, o_2, o_3\}$

$= \{o_5\}$

Handling of New and Assign

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)

Kind	Statement	Rule	PFG Edge
New	`i: x = new T()`	$\dfrac{}{o_i \in pt(x)}$	N/A
Assign	`x = y`	$\dfrac{o_i \in pt(\boldsymbol{y})}{o_i \in pt(\boldsymbol{x})}$	$x \leftarrow y$

Handling of Store and Load

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable `x` **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Kind	Statement	Rule	PFG Edge
Store	`x.f = y`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$	$o_i.f \leftarrow y$
Load	`y = x.f`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$	$y \leftarrow o_i.f$

Handling of Store and Load

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s) \rangle$ to WL

New points-to information may introduce new PFG edges

程序中存在 a.f=y c.f=y
那么就可能无需新建边，那怕第一次遇到

Kind	Statement	Rule	PFG Edge
Store	`x.f = y`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ o_j \in pt(\boldsymbol{y})}{o_j \in pt(\boldsymbol{o_i.f})}$	$o_i.f \leftarrow y$
Load	`y = x.f`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ o_j \in pt(\boldsymbol{o_i.f})}{o_j \in pt(\boldsymbol{y})}$	$y \leftarrow o_i.f$

Algorithms: Review

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable `x` **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

main algorithm

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S	Set of statements of the input program
WL	Work list
PFG	Pointer flow graph

An Example

```
Solve(S)
  WL = [], PFG = {}
  foreach i: x = new T() ∈ S do
    add ⟨x, {o_i}⟩ to WL

  foreach x = y ∈ S do
    AddEdge(y, x)

  while WL is not empty do
    remove ⟨n, pts⟩ from WL
    Δ = pts – pt(n)
    Propagate(n, Δ)
    if n represents a variable x then
      foreach o_i ∈ Δ do
        foreach x.f = y ∈ S do
          AddEdge(y, o_i.f)
        foreach y = x.f ∈ S do
          AddEdge(o_i.f, y)
```

S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: [ ]

PFG:

An Example

```
Solve(S)
  WL = [ ], PFG = {}
  foreach i: x = new T() ∈ S do
    add ⟨x, {o_i}⟩ to WL

  foreach x = y ∈ S do
    AddEdge(y, x)

  while WL is not empty do
    remove ⟨n, pts⟩ from WL
    Δ = pts – pt(n)
    Propagate(n, Δ)
    if n represents a variable x then
      foreach o_i ∈ Δ do
        foreach x.f = y ∈ S do
          AddEdge(y, o_i.f)
        foreach y = x.f ∈ S do
          AddEdge(o_i.f, y)
```

S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle b, \{o_1\}\rangle, \langle c, \{o_3\}\rangle]$

PFG:

An Example

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

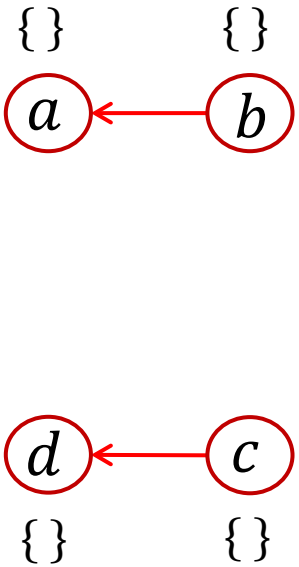
  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable `x` **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle b, \{o_1\}\rangle, \langle c, \{o_3\}\rangle]$

PFG:



An Example

Solve(S)
…
while WL is not empty **do**
remove $\langle n, pts \rangle$ from WL
$\Delta = pts - pt(n)$
Propagate(n, Δ)
if n represents a variable x **then**
foreach $o_i \in \Delta$ **do**
foreach `x.f = y` $\in S$ **do**
AddEdge($y, o_i.f$)
foreach `y = x.f` $\in S$ **do**
AddEdge($o_i.f, y$)

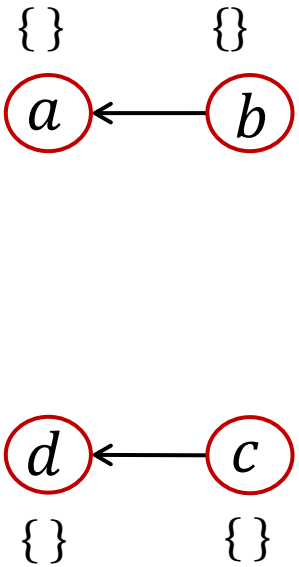
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle c, \{o_3\} \rangle]$

Processing: $\langle b, \{o_1\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

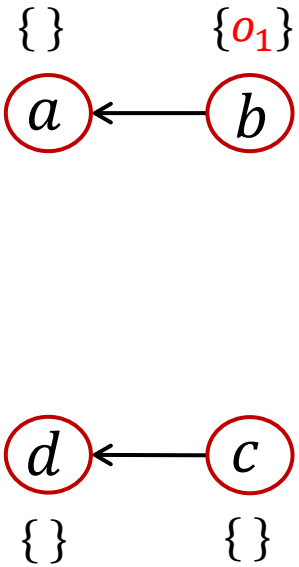
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle c, \{o_3\} \rangle]$

Processing: $\langle b, \{o_1\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

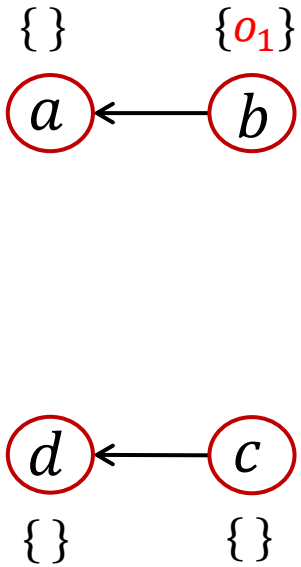
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle c, \{o_3\} \rangle, \langle a, \{o_1\} \rangle]$

Processing: $\langle b, \{o_1\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

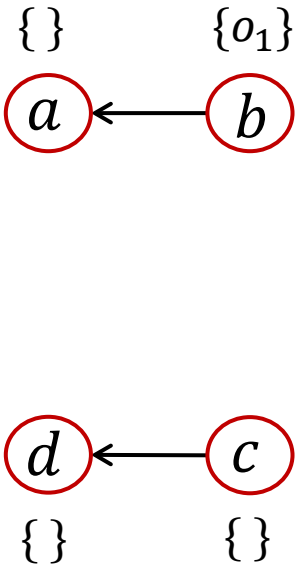
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle a, \{o_1\} \rangle]$

Processing: $\langle c, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

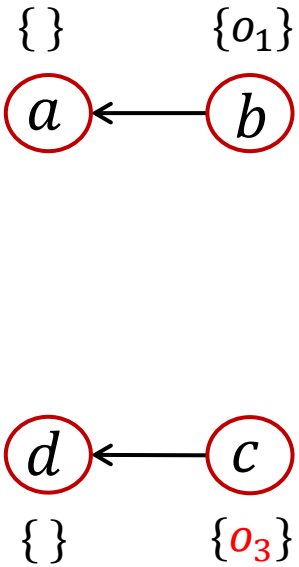
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle a, \{o_1\} \rangle]$

Processing: $\langle c, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

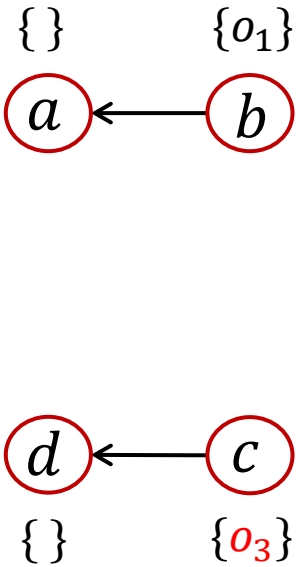
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle a, \{o_1\} \rangle, \langle d, \{o_3\} \rangle]$

Processing: $\langle c, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

S:

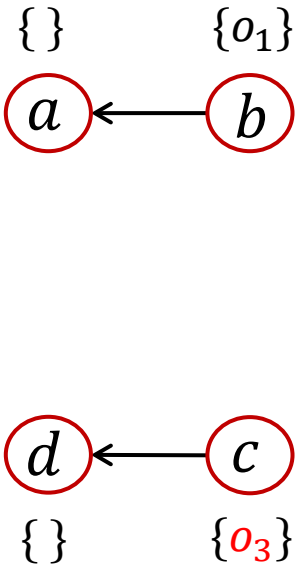
```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle a, \{o_1\} \rangle, \langle d, \{o_3\} \rangle]$

Processing: $\langle c, \{o_3\} \rangle$

What next?

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s) \rangle$ to WL

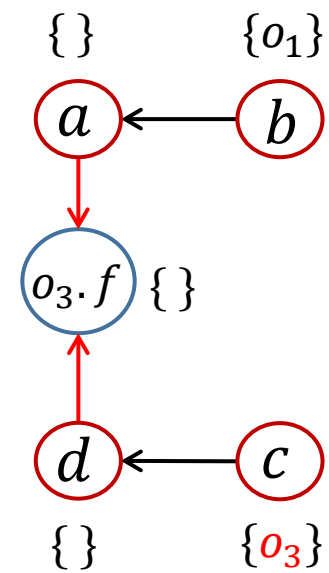
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle a, \{o_1\} \rangle, \langle d, \{o_3\} \rangle]$

Processing: $\langle c, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

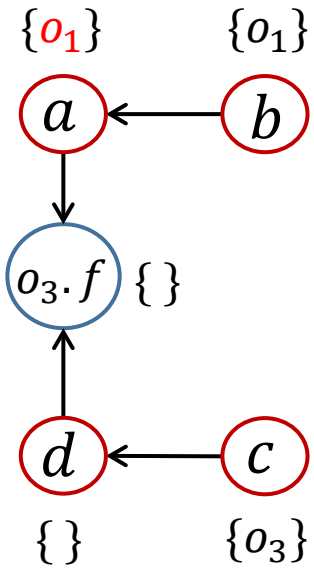
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle d, \{o_3\} \rangle]$

Processing: $\langle a, \{o_1\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

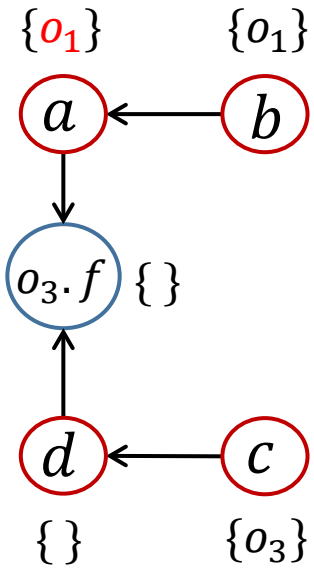
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle d, \{o_3\} \rangle, \langle o_3.f, \{o_1\} \rangle]$

Processing: $\langle a, \{o_1\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

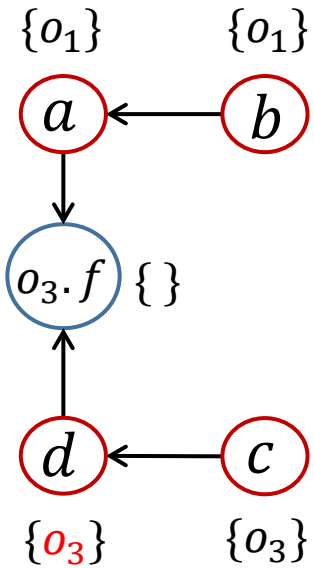
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_1\} \rangle]$

Processing: $\langle d, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

S:

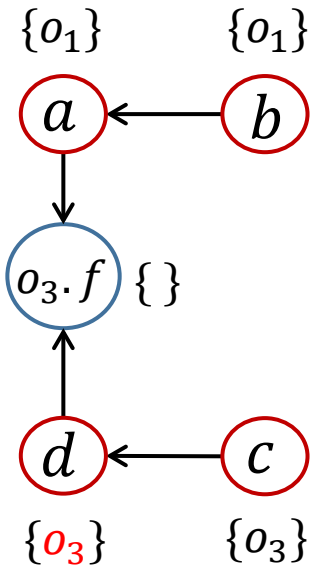
```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_1\} \rangle]$

Processing: $\langle d, \{o_3\} \rangle$

What next?

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

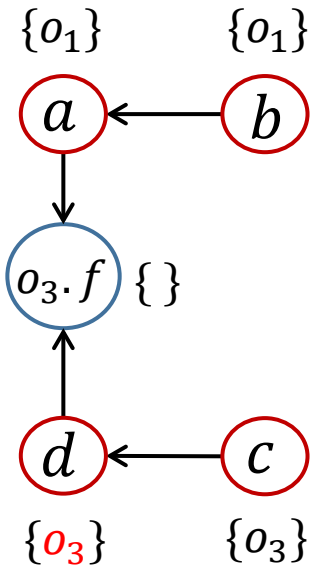
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_1\}\rangle, \langle o_3.f, \{o_3\}\rangle]$

Processing: $\langle d, \{o_3\}\rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S:

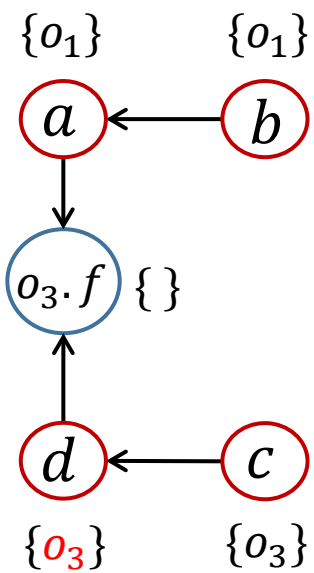
```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_1\}\rangle, \langle o_3.f, \{o_3\}\rangle]$

Processing: $\langle d, \{o_3\}\rangle$

What next?

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s) \rangle$ to WL

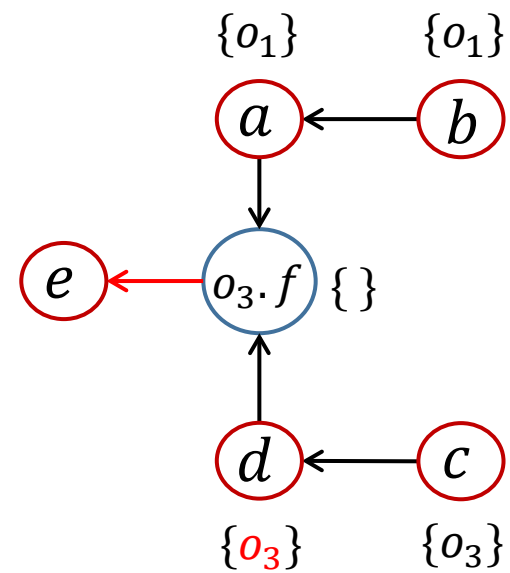
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_1\}\rangle, \langle o_3.f, \{o_3\}\rangle]$

Processing: $\langle d, \{o_3\}\rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

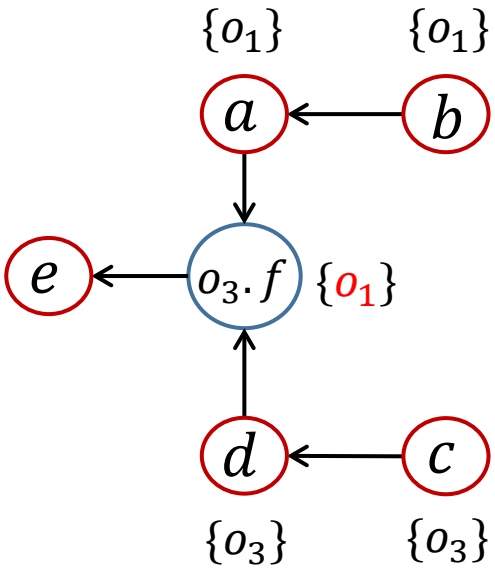
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle o_3.f, \{o_3\}\rangle, \langle e, \{o_1\}\rangle]$

Processing: $\langle o_3.f, \{o_1\}\rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

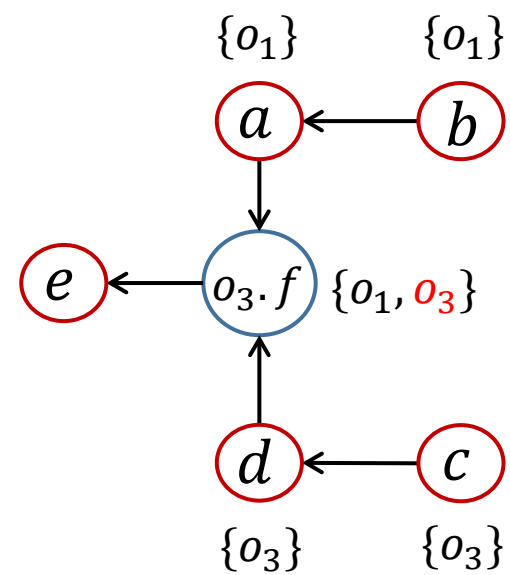
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: $[\langle e, \{o_1\} \rangle, \langle e, \{o_3\} \rangle]$

Processing: $\langle o_3.f, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

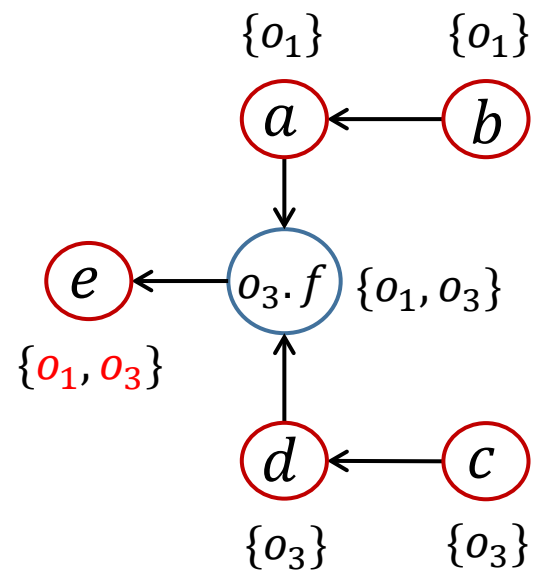
S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: [ ]

Processing: $\langle e, \{o_1\} \rangle, \langle e, \{o_3\} \rangle$

PFG:



An Example

Solve(S)
  …
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

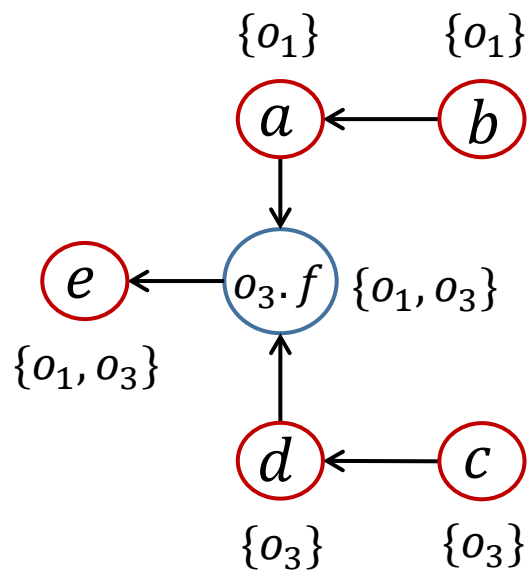
Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts \rangle$ to WL

S:

```
1 b = new C();
2 a = b;
3 c = new C();
4 c.f = a;
5 d = c;
6 c.f = d;
7 e = d.f;
```

WL: [ ]

PFG:



Algorithms: Review

Solve(S)
  $WL = [\,]$, $PFG = \{\}$
  foreach `i: x = new T()` $\in S$ **do**
    add $\langle x, \{o_i\}\rangle$ to WL

  foreach `x = y` $\in S$ **do**
    AddEdge(y, x)

  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge($y, o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f, y$)

main algorithm

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s)\rangle$ to WL

Propagate(n, pts)
  if pts is not empty **then**
    $pt(n) \cup= pts$
    foreach $n \rightarrow s \in PFG$ **do**
      add $\langle s, pts\rangle$ to WL

S	Set of statements of the input program
WL	Work list
PFG	Pointer flow graph

The X You Need To Understand in This Lecture

- Understand pointer analysis rules
- Understand pointer flow graph
- Understand pointer analysis algorithms

Static Program Analysis

Pointer Analysis Foundations (II)

Nanjing University

Tian Tan

2021

Contents

1. Pointer Analysis: Rules
2. How to Implement Pointer Analysis
3. Pointer Analysis: Algorithms
4. **Pointer Analysis with Method Calls**

Pointer Analysis in the Presence of Method Invocations

- Inter-procedural pointer analysis requires call graph

```
void foo(A a) {
    …
    b = a.bar();
    …
}
```

$pt(a) = ???$

$pt(b) = ???$

Pointer Analysis in the Presence of Method Invocations

- Inter-procedural pointer analysis requires call graph

CHA: resolve call targets based on declared type of a

```
void foo(A a) {
    …
    b = a.bar();
    …
}
```

$pt(a) = ???$

$pt(b) = ???$

- Call graph construction
  - ➢CHA: imprecise, introduce spurious call graph edges and points-to relations

Pointer Analysis in the Presence of Method Invocations

- Inter-procedural pointer analysis requires call graph

CHA: resolve call targets based on declared type of a

Pointer analysis: resolve call targets based on $pt(a)$

```
void foo(A a) {
    …
    b = a.bar();
    …
}
```

$pt(a) = ???$

$pt(b) = ???$

- Call graph construction
  - ➢CHA: imprecise, introduce spurious call graph edges and points-to relations
  - ➢Pointer analysis: more precise than CHA, both for call graph and points-to relations

a.k.a on-the-fly call graph construction

Rule: Call

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \mathrm{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$

Rule: Call

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ m = \text{Dispatch}(o_i, \mathrm{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$

- $\text{Dispatch}(o_i, \mathrm{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)

Rule: Call

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ m = \text{Dispatch}(o_i, \mathrm{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(\boldsymbol{m_{this}}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$

- $\text{Dispatch}(o_i, \mathrm{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m

Rule: Call

Kind	Statement	Rule
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(\boldsymbol{x}),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(\boldsymbol{m_{this}}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$

- $\text{Dispatch}(o_i, \text{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m

```
C x = new T();
…
r = x.foo(a1, a2);
```

```
class T … {   …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Kind	Statement	Rule
Call	l: r = x.k(a1,…,an)	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \mathrm{k}) \quad o_u \in pt(\boldsymbol{aj}), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(\boldsymbol{m_{pj}}), 1 \le j \le n \quad o_v \in pt(r)}$

- $\text{Dispatch}(o_i, \mathrm{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m
- m_{pj}: the *j*-th parameter of m

```
C x = new T();
…
r = x.foo(a1, a2);
```

```
class T … {   …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(\boldsymbol{aj}), 1 \le j \le n$ $o_v \in pt(m_{ret})$ ——— $o_i \in pt(m_{this})$ $o_u \in pt(\boldsymbol{m_{pj}}), 1 \le j \le n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$

- $\text{Dispatch}(o_i, \text{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m
- m_{pj}: the *j*-th parameter of m

```
C x = new T();
…
r = x.foo(a1, a2);
```

```
class T … {  …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Kind	Statement	Rule	PFG Edge
Call	l: `r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \mathrm{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(\boldsymbol{m_{ret}})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(\boldsymbol{r})}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$

1,2,3,4

- Dispatch(o_i, k): resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m
- m_{pj}: the *j*-th parameter of m
- m_{ret}: the variable that holds the return value of m

```
C x = new T();
…
r = x.foo(a1, a2);
```

```
class T … {  …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \le j \le n$ $o_v \in pt(\boldsymbol{m_{ret}})$ ——— $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \le j \le n$ $o_v \in pt(\boldsymbol{r})$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

- $\text{Dispatch}(o_i, \text{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m
- m_{pj}: the *j*-th parameter of m
- m_{ret}: the variable that holds the return value of m

```
C x = new T();
…
r = x.foo(a1, a2);

class T … {  …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ ——— $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

- $\text{Dispatch}(o_i, \text{k})$: resolves the virtual dispatch of k on o_i to a target method (based on type of o_i)
- m_{this}: this variable of m
- m_{pj}: the *j*-th parameter of m
- m_{ret}: the variable that holds the return value of m

```
C x = new T();
…
r = x.foo(a1, a2);

class T … {  …
  B foo(A p1, A p2) {
    this…
    return ret;
  }}
```

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

$pt(x) =$ { `new A`, `new B`, `new C` }

```
x.foo();
```

```
class A {
  T foo() {
    this…
  }}
```

```
class B extends A {
  T foo() {
    this…?
  }}
```

```
class C extends A {
  T foo() {
    this…
  }}
```

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ ——— $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

$pt(x) = \{$ `new A`, `new B`, `new C` $\}$

```
x.foo();
```

```
class A {
  T foo() {
    this…
  }}
```

```
class B extends A {
  T foo() {
    this…
  }}
```

```
class C extends A {
  T foo() {
    this…
  }}
```

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ $\overline{\qquad\qquad\qquad\qquad}$ $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

Receiver object should only flow to `this` variable of the **corresponding** target method

PFG edge $x \rightarrow m_{this}$ would introduce **spurious** points-to relations for `this` variables

$pt(x) = \{$ `new A`, `new B`, `new C` $\}$

```
x.foo();
```

这就是为什么不能直接连接 x 和 this的原因

```
class A {
  T foo() {
    this…
  }}
```
new A

```
class B extends A {
  T foo() {
    this…
  }}
```
new B

访问父property应该用supe

```
class C extends A {
  T foo() {
    this…
  }}
```
new C

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ $\overline{\qquad\qquad\qquad\qquad}$ $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

Receiver object should only flow to `this` variable of the **corresponding** target method

PFG edge $x \rightarrow m_{this}$ would introduce **spurious** points-to relations for `this` variables

$pt(x) = \{$ `new A`, `new B`, `new C` $\}$

`x.foo();`

With $x \rightarrow m_{this}$

```
class A {
  T foo() {
    this…
  }}
```

```
class B extends A {
  T foo() {
    this…
  }}
```

```
class C extends A {
  T foo() {
    this…
  }}
```

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ $\overline{\qquad\qquad\qquad\qquad}$ $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

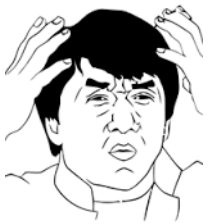
Receiver object should only flow to `this` variable of the **corresponding** target method

PFG edge $x \rightarrow m_{this}$ would introduce **spurious** points-to relations for `this` variables

$pt(x) =$ { new A, new B, new C }

```
x.foo();
```

With $x \rightarrow m_{this}$



```
class A {
  T foo() {
    this…
}}
```

{ new A, new B, new C }

```
class B extends A {
  T foo()
    this…
}}
```

{ new A, new B, new C }

```
class C extends A {
  T foo()
    this…
}}
```

{ new A, new B, new C }

Rule: Call

Why not add PFG edge $x \rightarrow m_{this}$?

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k})$ $o_u \in pt(aj), 1 \leq j \leq n$ $o_v \in pt(m_{ret})$ $\overline{\qquad\qquad\qquad\qquad}$ $o_i \in pt(m_{this})$ $o_u \in pt(m_{pj}), 1 \leq j \leq n$ $o_v \in pt(r)$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

Receiver object should only flow to `this` variable of the **corresponding** target method

PFG edge $x \rightarrow m_{this}$ would introduce **spurious** points-to relations for `this` variables

$pt(x) = \{$ `new A`, `new B`, `new C` $\}$

`x.foo();`

Without $x \rightarrow m_{this}$

```
class A {
  T foo() {
    this…
}}
```

{ new A }

```
class B extends A {
  T foo() {
    this…
}}
```

{ new B }

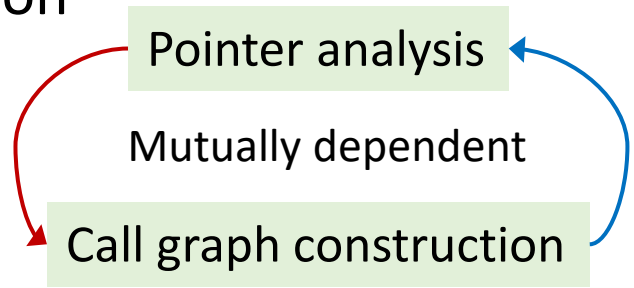
```
class C extends A {
  T foo() {
    this…
}}
```

{ new C }

Interprocedural Pointer Analysis

- Run together with call graph construction

```
void foo(A a) {
    …
    b = a.bar();
    …
}
```



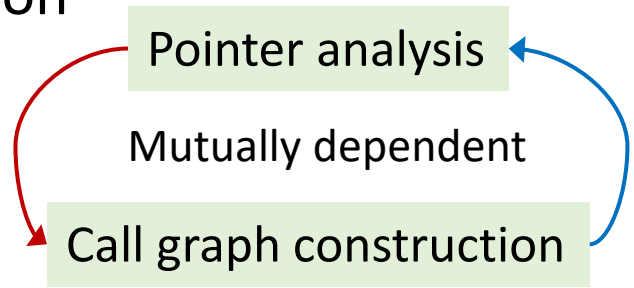
指针分析和 call graph 构建一起做

Interprocedural Pointer Analysis

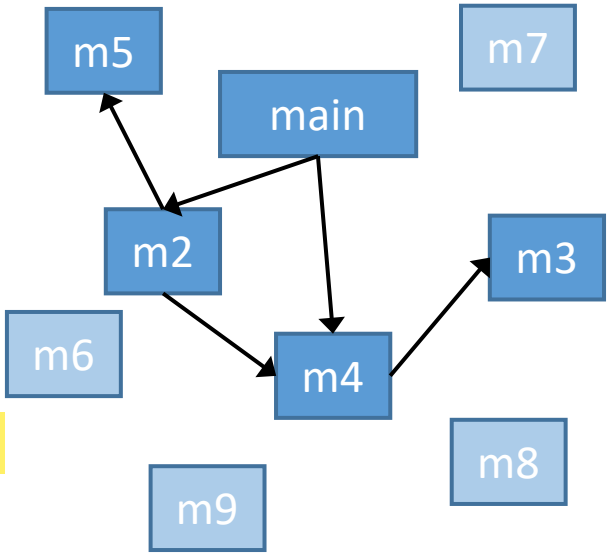
因为比如 a.bar这种
依赖于a的pointer sets，
才能决定dispatch哪个方法

- Run together with call graph construction

```
void foo(A a) {
    …
    b = a.bar();
    …
}
```



- Call graph forms a "reachable world"
  - Entry methods (e.g., the main method) are reachable from the beginning
  - The other reachable methods are gradually discovered during analysis
  - Only reachable methods and statements are analyzed



Algorithms

New part

```
Solve(m^entry)
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^entry)
  while WL is not empty do
    remove ⟨n, pts⟩ from WL
    Δ = pts – pt(n)
    Propagate(n, Δ)
    if n represents a variable x then
      foreach o_i ∈ Δ do
        foreach x.f = y ∈ S do
          AddEdge(y, o_i.f)
        foreach y = x.f ∈ S do
          AddEdge(o_i.f, y)
        ProcessCall(x, o_i)
```

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods
CG	Call graph edges

```
AddReachable(m)
  if m ∉ RM then
    add m to RM
    S ∪= S_m
    foreach i: x = new T() ∈ S_m do
      add ⟨x, {o_i}⟩ to WL
    foreach x = y ∈ S_m do
      AddEdge(y, x)
```

```
ProcessCall(x, o_i)
  foreach l: r = x.k(a1,…,an) ∈ S do
    m = Dispatch(o_i, k)
    add ⟨m_this, {o_i}⟩ to WL
    if l → m ∉ CG then
      add l → m to CG
      AddReachable(m)
      foreach parameter p_i of m do
        AddEdge(a_i, p_i)
      AddEdge(m_ret, r)
```

AddReachable(m)

- Expand the "reachable world", called
  - at the beginning for entry methods
  - when new call graph edge is discovered

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \mathrel{\cup}= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods

AddReachable(m)

- Expand the "reachable world", called
  - at the beginning for entry methods
  - when new call graph edge is discovered

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \ \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

Add new reachable method and statements

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods

AddReachable(m)

- Expand the "reachable world", called
  - ➢ at the beginning for entry methods
  - ➢ when new call graph edge is discovered

AddEdge(s, t)
  if $s \rightarrow t \notin PFG$ **then**
    add $s \rightarrow t$ to PFG
    if $pt(s)$ is not empty **then**
      add $\langle t, pt(s) \rangle$ to WL

(Same as before)

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \mathrel{\cup}= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\} \rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

Add new reachable method and statements

Update worklist and PFG for new discovered statements

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods

为什么刚开始不处理 load 和 store呢?

因为 x.f =y y.f=x 均是需要知道x指向的o才行

这个算法是 tantian 自创的！！！！

Algorithms

New part

Solve(m^{entry})
  WL=[ ], PFG={}, S={}, RM={}, CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods
CG	Call graph edges

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

为什么不直接添加 l -> m ?

因为 oj，ok 等可能跟 oi是一个类型， 导致已经添加过了

argument 实参、 parameter 形参

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

ProcessCall(x, o_i)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

Pass receiver object to *this* variable

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

ProcessCall(x, o_i)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

Pass receiver object to *this* variable

Construct call graph on the fly

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

ProcessCall(x, o_i)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

Pass receiver object to *this* variable

Construct call graph on the fly

Pass arguments

Pass return values

Kind	Statement	Rule	PFG Edge
Call	`l: r = x.k(a1,…,an)`	$\dfrac{o_i \in pt(x),\ m = \text{Dispatch}(o_i, \text{k}) \quad o_u \in pt(aj), 1 \le j \le n \quad o_v \in pt(m_{ret})}{o_i \in pt(m_{this}) \quad o_u \in pt(m_{pj}), 1 \le j \le n \quad o_v \in pt(r)}$	$a1 \rightarrow m_{p1}$ … $an \rightarrow m_{pn}$ $r \leftarrow m_{ret}$

Algorithms Output: Points-to Relations (pt) Call Graph (CG)

Solve(m^{entry})
  WL=[ ], PFG={}, S={}, RM={}, CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

S	Set of reachable statements
S_m	Set of statements in method m
RM	Set of reachable methods
CG	Call graph edges

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\} \rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\} \rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

An Example

右侧应该是ir

不过这里简化理解了

然后ir里
会被处理成
<A,A.foo,A>
<A,B.foo,A>这种形式的

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

An Example

Solve(m^{entry})
WL=[ ],PFG={},S={},RM={},CG={}
AddReachable(m^{entry})
while WL is not empty **do**
remove $\langle n, pts \rangle$ from WL
$\Delta = pts - pt(n)$
Propagate(n, Δ)
if n represents a variable x **then**
foreach $o_i \in \Delta$ **do**
foreach `x.f = y` $\in S$ **do**
AddEdge(y, $o_i.f$)
foreach `y = x.f` $\in S$ **do**
AddEdge($o_i.f$, y)
ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

WL: [ ]

RM: { }

CG: { }

PFG:

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

WL: [ ]

RM: { }

CG: { }

PFG:

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

```
class A {
  static void main() {
    A a = new A();
    A b = new B();
    A c = b.foo(a);
  }
  A foo(A x) { … }
}
class B extends A {
  A foo(A y) {
    A r = new A();
    return r;
  }
}
```

WL: [ ]

RM: { A.main() }

CG: { }

PFG:

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

WL: $[\langle a, \{o_3\}\rangle, \langle b, \{o_4\}\rangle]$

RM: { A.main() }

CG: { }

PFG:

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

WL: $[\langle b, \{o_4\} \rangle]$

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle a, \{o_3\} \rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_3\}$

a

WL: $[\langle b, \{o_4\} \rangle]$

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle a, \{o_3\} \rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_3\}$

a

WL: [ ]

RM: { A.main() }

CG: { }

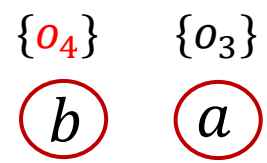
PFG:

Processing:
$\langle b, \{o_4\} \rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```



WL: [ ]

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle b, \{o_4\} \rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

What next?

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

WL: [ ]

RM: { A.main() }

Processing:
$\langle b, \{o_4\} \rangle$

CG: { }

PFG:

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

→ ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

WL: [ ]

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

m = ?

ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

WL: [ ]

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

m = B.foo(A)

ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

WL: [ ]

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

ProcessCall(x, o_i)

WL: [$\langle$B.foo/$this$, $\{o_4\}\rangle$]

RM: { A.main() }

CG: { }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

ProcessCall(x, o_i)

WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle]$

RM: { A.main() }

CG: { 5 → B.foo(A) }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

$\{o_4\}$ $\{o_3\}$

b a

WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle]$

RM: { A.main() }

Processing:
$\langle b, \{o_4\}\rangle$

CG: { 5 → B.foo(A) }

PFG:

CHA: { 5 → B.foo(A) , 5 → A.foo(A) }

Spurious call edge

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

What change?

ProcessCall(x, o_i)

$\{o_4\}$ $\{o_3\}$

b a

WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle]$

RM: { A.main() }

CG: { 5 → B.foo(A) }

PFG:

Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

ProcessCall(x, o_i)

WL: [$\langle$B.foo/$this$, $\{o_4\}\rangle$]

RM: { A.main() }

CG: { 5 $\rightarrow$ B.foo(A) }

Processing:
$\langle b, \{o_4\}\rangle$

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

An Example

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

Processing:
$\langle b, \{o_4\}\rangle$

An Example

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

ProcessCall(x, o_i)
  foreach `l: r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

AddReachable(m)
  if $m \notin RM$ **then**
    add m to RM
    $S \cup= S_m$
    foreach `i: x = new T()` $\in S_m$ **do**
      add $\langle x, \{o_i\}\rangle$ to WL
    foreach `x = y` $\in S_m$ **do**
      AddEdge(y, x)

WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle, \langle r, \{o_{11}\}\rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

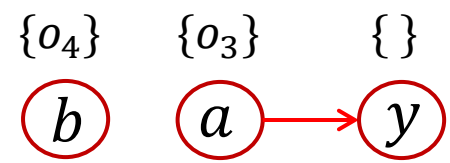
Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
      AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```



WL: $[\langle \text{B.foo}/this, \{o_4\}\rangle, \langle r, \{o_{11}\}\rangle, \langle y, \{o_3\}\rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

PFG:

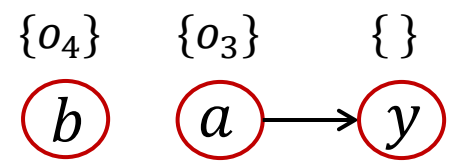
Processing:
$\langle b, \{o_4\}\rangle$

An Example

ProcessCall(x, o_i)
  foreach l: `r = x.k(a1,…,an)` $\in S$ **do**
    m = Dispatch(o_i, k)
    add $\langle m_{this}, \{o_i\}\rangle$ to WL
    if $l \rightarrow m \notin CG$ **then**
      add $l \rightarrow m$ to CG
      AddReachable(m)
      foreach parameter p_i of m **do**
        AddEdge(a_i, p_i)
    AddEdge(m_{ret}, r)

ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

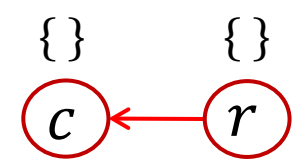


WL: [⟨B.foo/$this$, $\{o_4\}$⟩, ⟨r, $\{o_{11}\}$⟩, ⟨y, $\{o_3\}$⟩]

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

PFG:

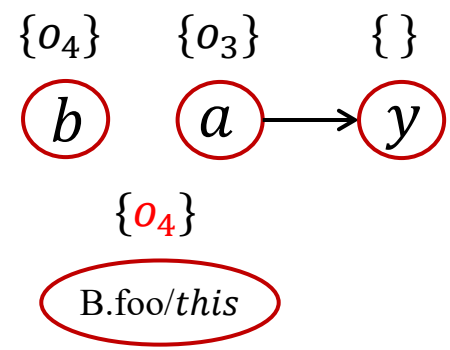


Processing:
$\langle b, \{o_4\}\rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

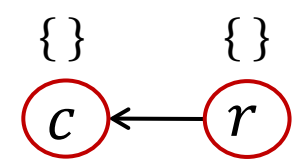


WL: $[\langle r, \{o_{11}\}\rangle, \langle y, \{o_3\}\rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

PFG:

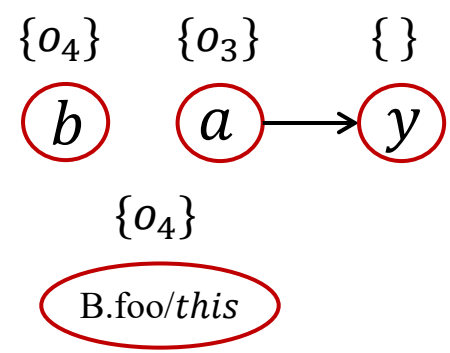


Processing:
$\langle$B.foo/$this$, $\{o_4\}\rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts\rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

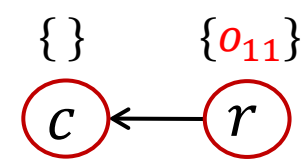


WL: $[\langle y, \{o_3\}\rangle, \langle c, \{o_{11}\}\rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

PFG:

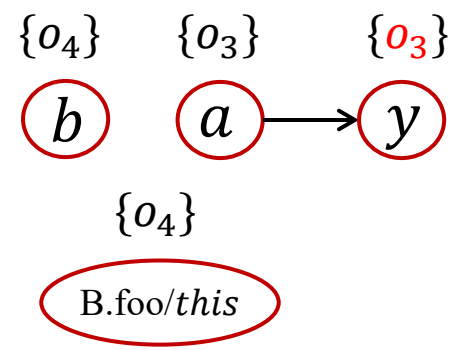


Processing:
$\langle r, \{o_{11}\}\rangle$

An Example

Solve(m^{entry})
  WL=[ ], PFG={}, S={}, RM={}, CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

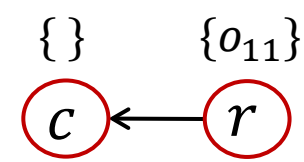


WL: $[\langle c, \{o_{11}\} \rangle]$

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

PFG:

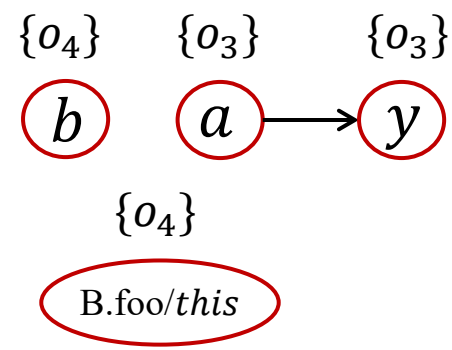


Processing:
$\langle y, \{o_3\} \rangle$

An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```



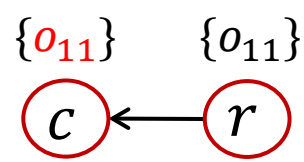
WL: [ ]

RM: { A.main(), B.foo(A) }

CG: { 5 → B.foo(A) }

Processing:
$\langle c, \{o_{11}\} \rangle$

PFG:

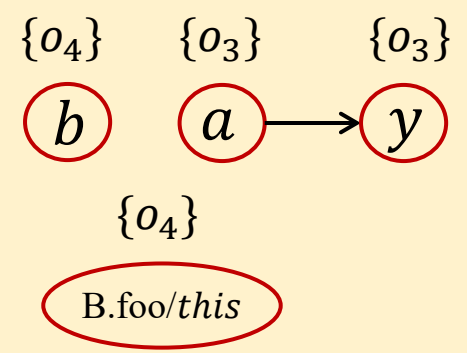


An Example

Solve(m^{entry})
  WL=[ ],PFG={},S={},RM={},CG={}
  AddReachable(m^{entry})
  while WL is not empty **do**
    remove $\langle n, pts \rangle$ from WL
    $\Delta = pts - pt(n)$
    Propagate(n, Δ)
    if n represents a variable x **then**
      foreach $o_i \in \Delta$ **do**
        foreach `x.f = y` $\in S$ **do**
          AddEdge(y, $o_i.f$)
        foreach `y = x.f` $\in S$ **do**
          AddEdge($o_i.f$, y)
        ProcessCall(x, o_i)

```
1  class A {
2    static void main() {
3      A a = new A();
4      A b = new B();
5      A c = b.foo(a);
6    }
7    A foo(A x) { … }
8  }
9  class B extends A {
10   A foo(A y) {
11     A r = new A();
12     return r;
13   }
14 }
```

Algorithm finishes



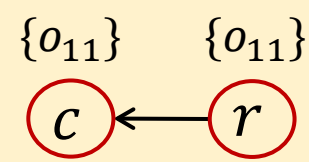
WL: [ ]

RM: { A.main(), B.foo(A) }

Processing:

CG: { 5 → B.foo(A) }

PFG:



Final results

The X You Need To Understand in This Lecture

- Understand pointer analysis rule for method call
- Understand inter-procedural pointer analysis algorithm
- Understand on-the-fly call graph construction